

TAZEWELL COUNTY AP, VA, USA

WMO: 720264

Lat: 37.064N

Lon: 81.798W

Elev: 2652

StdP: 13.34

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	12.1	-11.1	3.3	35.9	-3.3	5.1	28.4	19.4	36.1	17.6	36.3	6.4	300	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	86.3	71.1	83.9	70.0	81.8	69.0	74.8	81.9	73.2	79.8	72.0	78.4	4.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	133.3	78.5	71.8	129.6	77.5	70.2	122.6	75.6	40.2	82.2	38.8	80.4	37.6	77.3	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.5	12.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-0.3	75.9	3.7	2.5	-2.9	77.7	-5.1	79.2	-7.1	80.6	-9.8	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	33.5	37.2	43.3	54.4	62.6	69.6	72.3	71.9	67.2	56.2	46.0	37.5		
	DBStd	16.35	12.16	11.47	10.44	9.91	7.69	4.70	3.69	4.00	6.00	8.89	9.06	10.83		
	HDD50	1870	522	377	260	69	7	0	0	0	0	45	178	412		
	HDD65	4679	976	779	672	332	138	15	2	3	46	290	572	854		
	CDD50	3472	11	19	54	202	397	587	690	679	515	238	57	23		
	CDD65	801	0	1	0	14	62	152	227	216	110	18	1	0		
	CDH74	4443	0	0	0	64	306	849	1282	1228	632	81	1	0		
	CDH80	922	0	0	0	3	31	176	288	287	127	10	0	0		
(14) Wind	WSAvg	5.0	6.4	6.8	6.7	6.7	4.4	3.8	3.2	3.0	3.6	4.7	5.3	5.7		
(15) Precipitation	PrecAvg	44.0	3.5	3.3	4.4	3.6	4.5	4.3	5.0	4.0	3.0	2.5	2.8	3.4		
	PrecMax	56.1	6.3	7.9	7.3	6.9	8.3	7.2	9.9	6.7	5.9	5.2	6.1	7.2		
	PrecMin	33.9	1.5	1.0	1.7	1.5	2.0	2.0	2.2	1.8	0.9	0.5	0.6	1.3		
	PrecStd	6.7	1.3	1.5	1.7	1.4	1.9	1.5	1.6	1.3	1.1	1.2	1.3	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.2	69.2	72.1	79.3	83.3	88.2	89.8	89.6	87.6	80.9	70.2	65.7		
		MCWB	56.3	59.3	54.1	59.9	67.1	70.7	73.8	73.5	66.9	63.2	54.3	55.6		
	2%	DB	58.7	62.6	66.3	75.2	80.5	84.0	85.8	85.9	83.5	75.4	65.6	61.1		
		MCWB	53.0	54.0	50.6	57.3	65.1	69.1	72.1	72.0	66.9	61.1	49.7	53.6		
	5%	DB	54.7	57.3	63.2	72.6	77.6	81.7	82.7	83.0	81.1	72.3	62.9	57.3		
		MCWB	49.2	50.8	50.8	56.6	63.4	68.3	70.9	71.0	66.1	59.2	52.0	52.0		
	10%	DB	51.8	53.9	59.7	69.6	74.8	79.3	81.1	81.1	77.5	69.4	59.5	54.3		
		MCWB	47.2	47.2	48.0	56.0	62.3	67.1	70.5	69.9	65.2	58.6	47.7	48.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	59.4	58.2	63.3	70.3	74.8	76.6	76.8	73.2	68.0	60.4	58.6		
		MCDB	61.0	66.5	66.3	73.1	79.6	81.7	84.7	83.6	79.8	75.5	66.0	62.5		
	2%	WB	54.1	56.7	55.7	60.9	68.0	72.7	75.1	74.6	70.6	65.8	57.3	56.3		
		MCDB	57.6	61.1	62.2	70.6	76.6	80.3	82.4	81.8	77.6	72.0	61.4	59.4		
	5%	WB	51.0	52.9	53.7	59.3	66.5	71.2	73.4	73.0	69.0	63.4	54.7	52.7		
		MCDB	54.0	56.6	59.9	68.6	74.4	78.6	79.7	79.7	76.3	68.3	59.2	55.9		
	10%	WB	47.3	48.4	50.4	57.4	64.6	69.7	72.2	71.5	67.5	61.0	52.0	49.7		
		MCDB	49.9	52.2	57.6	66.2	71.8	76.7	78.0	78.1	74.9	66.4	56.7	54.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	16.0	16.8	18.4	17.7	17.3	16.5	16.5	17.1	16.5	16.5	13.4		
		MCDBR	16.7	17.5	19.8	19.9	19.4	20.0	19.4	19.3	20.7	19.6	18.3	13.9		
		MCWBR	12.8	12.6	12.7	10.3	9.2	9.3	8.3	8.4	8.2	9.8	11.7	10.6		
	5% WB	MCDBR	15.7	15.4	17.7	17.0	18.1	17.7	16.9	16.5	17.2	15.6	15.8	14.4		
		MCWBR	13.9	13.2	13.4	10.5	9.6	9.1	8.5	8.1	8.1	9.5	12.8	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.295	0.317	0.356	0.393	0.421	0.449	0.454	0.358	0.331	0.299	0.282		
	taud		2.536	2.503	2.485	2.386	2.276	2.228	2.184	2.143	2.407	2.491	2.555	2.572		
	Ebn at Noon		291	300	302	293	282	273	264	260	283	283	283	285		
	Edh at Noon		24	29	32	37	43	44	46	47	34	29	24	22		
(44) All-Sky Solar Radiation	RadAvg		670	868	1230	1585	1757	1914	1784	1672	1423	1095	809	589		
	RadStd		61	112	92	121	135	123	113	83	130	135	86	60		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+148

Nomenclature: See separate page